

#6 Remote Controlled Smart Home System

Objective

To design a Remote-Controlled Smart Home System

Description

This project uses an STM32F405RGT6 development board to demonstrate the functionality of a Remote-Controlled Smart Home System. The system integrates LED Bulbs and a DC motor to simulate home appliances like lights and fans. The appliances can be controlled via UART commands sent through a PC, with their statuses displayed on an LCD. Manual control using push buttons is also implemented.

Hardware Required

1. CORTEX M4-STM32F405 Board
2. DC LED Bulb – Simulates the room light
3. DC Motor – Simulates a ceiling fan or any rotary appliance
4. Push Buttons – 3Nos
5. IR Remote
6. IR Receiver Module
7. Motor Driver - Dual TB6612FNG
8. Standard 5V 2A Power Supply with Dual Pin DC Plug
9. DC Power Module DC Power Adapter Plate
10. 12V Battery
11. Relay module
12. LED - Red

Software Required

1. STM32 CUBE IDE

Expected Result & Test Cases

1. Controlling Lights & FAN using UART Messages

- Sending "Light ON" via UART turns on the Light
- Sending "Light OFF" via UART turns off the Light
- Sending "Fan ON" via UART starts the FAN at 10% Speed (Default Speed)
- Sending "Fan OFF" via UART stops the FAN
- Sending "Fan Speed 1" via UART starts the FAN at 30% Speed
- Sending "Fan Speed 2" via UART starts the FAN at 50% Speed

- Sending "Fan Speed 3" via UART starts the FAN at 80% Speed

2. **Controlling Lights & FAN using Remote Control**

- Just Press RED "Power" Button – Turns On/Off the light.
- Just Press GREEN "Play/Pause" Button – Starts/Stops the Fan. (Default Speed)
- Just Press Button "1" – The Fan rotates at 30% Speed.
- Just Press Button "2" – The Fan rotates at 50% Speed.
- Just Press Button "3" – The Fan rotates at 80% Speed.

3. **Manual Override:**

- One button is assigned to Light and another to Fan.
- Pressing the corresponding button will toggle the corresponding device.

4. **Status Display:**

- The LCD continuously updates the appliance statuses.
- **Ex:** Light Status: ON (LCD Line #1)
- Fan Status: OFF (LCD Line #2)

Test Conditions

- **Handling invalid commands via UART:**
 - When an invalid command (e.g., "InvalidCmd") is sent via UART, the LCD should display "Error: Invalid Command," and a red LED should blink to indicate the error.
- **Test for Wrong Sequence of commands via UART:**
 - Trying to change the Fan Speed without turning On the Fan. Display appropriate error message on the LCD display.
- **System reinitialization on reset or power cycle:**
 - Simulate the system reset using a push button, all appliances should be turned off by default, and the LCD should reinitialize to show the default status of all devices.

Project Deliverables

1. Project Report containing
 - i.Flowchart
 - ii.H/w Circuit - Photos
 - iii.Code
 - iv.Output Screenshots/Photos
 - v.Live Expressions - Screenshots

2. Project Report to be Uploaded into LMS

Project Assessment

Project Activities	Weightage (%)
H/w Setup	25%
Coding	25%
Testing	25%
Documentation	25%